THE COTANGENT COMPLEX OF A LOCAL HOMOMORPHISM IS UNCHANGED BY HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays both chains of equalities; the proof treats the henselization case and states that the strict case is similar, with the same two triangles available.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The vanishing of the cotangent complex of a henselization over the ring. Lemma lemma-when-zero, used at proof lines 1905-1907. Kills the four comparison complexes.
- The fundamental distinguished triangle for a composition of ring maps. equation-triangle, used at proof lines 1908-1912. Produces the two identifications from which the chain follows.
- The definition of the cotangent complex of a ring map. used at proof lines 1896-1902. Fixes the objects appearing in the displayed chains.

Source reference. Original equation 08QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11070–11070. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08R2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11080–11080. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $A \to B$ be a local ring homomorphism of local rings. Let $A^h \to B^h$, resp. $A^{sh} \to B^{sh}$ be the induced maps of henselizations, resp. strict henselizations. Then*

$$L_{B^h/A^h} = L_{B^h/A} = L_{B/A} \otimes^{\mathbf{L}}_B B^h \quad \text{resp.} \quad L_{B^{sh}/A^{sh}} = L_{B^{sh}/A} = L_{B/A} \otimes^{\mathbf{L}}_B B^{sh}$$

in $D(B^h)$, resp. $D(B^{sh})$.

Proof. The complexes $L_{A^h/A}$, $L_{A^{sh}/A}$, $L_{B^h/B}$, and $L_{B^{sh}/B}$ are all zero by Lemma 08R2. Using the fundamental distinguished triangle (08QS) for $A \to B \to B^h$ we

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obtain $L_{B^h/A} = L_{B/A} \otimes^{\mathbf{L}}_B B^h$. Using the fundamental triangle for $A \to A^h \to B^h$ we obtain $L_{B^h/A^h} = L_{B^h/A}$. Similarly for strict henselizations. $\square$